

BALLINA, AustraliaWMO: **945960**Lat: **28.8353S**Lon: **153.5586E**Elev: **7**StdP: **14.69**Time Zone: **10.00 (AUS)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	42.1	44.4	32.0	26.5	57.1	35.0	29.8	56.3	23.3	62.2	20.9	62.7	1.5	280	0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.1	87.4	74.8	84.8	74.1	82.8	73.0	77.5	84.0	76.2	82.3	74.9	80.6	14.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.6	134.2	80.8	74.3	128.2	79.7	73.0	122.7	78.7	41.0	84.6	39.7	82.3	38.5	80.9	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.2	19.9	18.1	37.4	94.9	1.6	4.5	36.3	98.1	35.3	100.8	34.4	103.3	33.3	106.6		
(5)			36.2	80.2	2.0	2.1	34.8	81.7	33.6	83.0	32.5	84.2	31.1	85.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.7	74.7	74.5	72.7	68.0	62.8	59.3	57.6	58.9	63.1	66.6	69.7	72.8
	DBStd	7.03	3.40	3.11	3.00	3.10	3.68	3.48	3.58	3.90	4.14	4.01	4.21	3.84
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	810	0	0	0	8	87	171	231	195	84	28	6	0
	CDD50	6089	765	686	703	540	397	280	237	276	394	514	591	706
	CDD65	1425	300	266	238	98	19	1	2	6	29	78	147	241
	CDH74	7999	1981	1704	1184	325	54	2	11	58	159	375	723	1423
	CDH80	1333	387	306	141	20	2	0	0	9	21	54	128	265
Wind	WSAvg	8.5	10.0	9.6	8.4	7.8	6.8	7.4	7.2	7.7	8.6	9.4	9.8	9.7
Precipitation	PrecAvg	66.4	5.8	7.7	8.0	7.0	6.9	5.6	4.1	3.7	2.4	4.7	4.6	5.8
	PrecMax	104.5	18.7	28.0	24.8	18.7	20.7	14.5	15.0	8.1	7.4	25.0	11.1	18.5
	PrecMin	35.8	1.0	1.8	0.9	1.1	0.7	0.8	0.2	0.2	0.0	0.2	1.0	1.0
	PrecStd	18.0	4.3	5.5	6.0	4.6	4.6	4.1	3.9	2.6	2.1	5.1	2.6	4.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.7	90.0	87.4	82.7	78.6	73.7	75.5	79.2	82.6	85.0	87.9	89.6
		MCWB	75.7	77.6	76.0	71.5	65.9	62.8	62.5	61.2	64.9	67.1	71.9	75.8
	2%	DB	86.4	85.7	83.3	79.3	75.1	71.0	71.6	74.7	78.1	81.0	83.5	85.8
		MCWB	75.6	76.1	73.9	69.2	64.2	61.6	59.7	60.8	64.8	68.7	71.7	74.3
	5%	DB	83.7	83.3	81.2	77.4	72.9	69.0	68.7	71.3	75.2	78.0	80.2	82.8
		MCWB	74.5	74.5	72.2	68.3	63.6	60.8	59.3	59.5	64.2	67.3	70.6	72.8
	10%	DB	81.8	81.4	79.6	75.7	71.0	67.1	66.3	68.6	72.6	75.4	77.8	80.5
		MCWB	73.3	73.0	71.2	67.3	63.1	60.2	58.3	58.7	62.9	66.4	69.2	71.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.4	79.7	77.4	74.2	69.8	66.4	66.6	66.3	69.3	72.8	75.6	78.4
		MCDB	84.7	86.2	84.3	79.3	74.0	69.7	71.7	74.3	77.0	79.8	83.8	86.0
	2%	WB	77.2	77.6	75.4	71.4	67.8	64.5	63.9	63.7	67.6	70.8	73.4	76.1
		MCDB	83.6	83.9	80.9	76.4	72.2	68.0	67.9	69.8	74.6	77.4	80.4	83.0
	5%	WB	75.7	76.1	74.1	69.9	65.9	63.0	61.6	62.1	66.2	69.2	71.8	74.5
		MCDB	81.7	81.6	79.1	75.3	70.6	66.8	66.0	68.1	72.6	75.3	77.8	80.8
	10%	WB	74.5	74.7	72.8	68.6	64.4	61.7	59.8	60.7	64.8	67.7	70.5	72.9
		MCDB	80.1	79.7	77.7	74.0	69.1	65.6	64.3	66.9	70.7	73.7	76.2	78.6
Mean Daily Temperature Range	5% DB	MDBR	13.1	13.0	13.4	14.8	16.7	15.6	18.2	20.0	18.9	16.3	15.0	14.3
		MCDBR	16.3	16.2	16.3	17.7	19.6	18.4	22.6	25.3	22.6	21.2	18.0	17.6
		MCWBR	7.7	8.0	8.1	9.1	10.5	10.5	12.6	13.6	11.9	11.2	8.6	8.5
		MCDDBR	14.6	14.4	14.1	15.2	15.6	14.4	16.7	20.3	18.8	17.4	16.1	15.2
	5% WB	MCWBR	8.0	8.0	7.7	8.7	9.3	9.5	11.2	12.3	11.0	10.2	8.4	8.1
Clear-Sky Solar Irradiance	taub	0.389	0.374	0.357	0.344	0.319	0.313	0.304	0.317	0.354	0.384	0.382	0.392	
	taud	2.507	2.561	2.607	2.614	2.645	2.650	2.663	2.606	2.499	2.443	2.490	2.487	
	Ebn at Noon	302	301	297	285	279	274	282	290	293	295	302	302	
	Edh at Noon	36	33	30	27	24	23	23	27	33	37	36	37	
All-Sky Solar Radiation	RadAvg	2061	1874	1586	1332	1098	908	1055	1338	1668	1869	2031	2045	
	RadStd	162	146	87	67	66	58	84	101	130	161	161	203	